

Should-Cost & Make-vs-Buy Report

cube.step — 2026-07-10 · material class: polymer

Assumption-based should-cost. Every figure is a glass-box estimate built from stated drivers and rate assumptions (tagged DEFAULT / SHOP / USER below). Confidence bands are **assumption-based, not yet validated** against measured shop outcomes — they are propagated from the stated assumptions, not a guarantee. Use as a negotiation and design baseline, not a quote.

Geometry

Volume (cm³)	Surface area (cm²)	Bounding box (mm)	Watertight	Faces
2.72	14.32	20.0 × 15.0 × 10.0	Yes	185,308

Geometric Routing

Recommended process: **mjf** (archetype: bulk_solid, confidence 0.4)

General solid (91% of bbox filled, ~3.8mm mean wall) with no dominant sheet/rotational/prismatic signature → additive (MJF/SLS) for a polymer prototype.

Per-Process Estimates (should-cost, USD)

fdm / PLA — qty 1 — \$30.00/unit

Line items

Cost element	USD/unit
labor	\$7.05
machine	\$1.41
material	\$0.09
amortized_fixed	\$8.75
min_charge_floor	\$12.70
Σ unit cost	\$30.00

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0927 \$	CAD volume 2.72 cm³ × PLA density 1.24 g/cm³ = 0.0034 kg × \$25/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1	DEFAULT
parts_per_build	68.0 parts	XY nest: plate 250×250mm × xy_packing 0.5 ÷ footprint (15.0×20.0+4mm) = 68 parts/plate	DEFAULT
machine_cost	1.4057 \$	0.1757 hr × \$8/hr × region-labor ×1 [serial XY-nested: deposition V/16 = 2.72/16 = 0.170hr/part (per-part nozzle) + Z-sweep (10.0/25)÷68 parts/plate = 0.006hr/part (plate Z-climb amortized; XY packing 0.5, plate 250×250mm) = 0.176hr/part]	DEFAULT
cycle_time	0.1757 hr	serial XY-nested: deposition V/16 = 2.72/16 = 0.170hr/part (per-part nozzle) + Z-sweep (10.0/25)÷68 parts/plate = 0.006hr/part (plate Z-climb amortized; XY packing 0.5, plate 250×250mm) = 0.176hr/part	DEFAULT
labor_cost	7.0515 \$	finish 0.2hr/part + bulk 0.1hr/build ÷ 68 = 0.201hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	8.75 \$	setup 0.25hr × \$35/hr × ceil(1/68) = 1 setups ÷ 1 × region-labor ×1	DEFAULT
min_charge_floor	12.7001 \$	shop/order minimum \$30/lot × 1 lots ÷ 1 = \$30.00/unit floor (applied)	DEFAULT

Confidence 80%: \$18.00–\$42.00/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

fdm / PLA — qty 100 — \$8.72/unit

Line items

Cost element	USD/unit
labor	\$7.05
machine	\$1.41
material	\$0.09
amortized_fixed	\$0.17
Σ unit cost	\$8.72

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0927 \$	CAD volume 2.72 cm³ × PLA density 1.24 g/cm³ = 0.0034 kg × \$25/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1	DEFAULT
parts_per_build	68.0 parts	XY nest: plate 250×250mm × xy_packing 0.5 ÷ footprint (15.0×20.0+4mm) = 68 parts/plate	DEFAULT
machine_cost	1.4057 \$	0.1757 hr × \$8/hr × region-labor ×1 [serial XY-nested: deposition V/16 = 2.72/16 = 0.170hr/part (per-part nozzle) + Z-sweep (10.0/25)÷68 parts/plate = 0.006hr/part (plate Z-climb amortized; XY packing 0.5, plate 250×250mm) = 0.176hr/part]	DEFAULT
cycle_time	0.1757 hr	serial XY-nested: deposition V/16 = 2.72/16 = 0.170hr/part (per-part nozzle) + Z-sweep (10.0/25)÷68 parts/plate = 0.006hr/part (plate Z-climb amortized; XY packing 0.5, plate 250×250mm) = 0.176hr/part	DEFAULT
labor_cost	7.0515 \$	finish 0.2hr/part + bulk 0.1hr/build ÷ 68 = 0.201hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	0.175 \$	setup 0.25hr × \$35/hr × ceil(100/68) = 2 setups ÷ 100 × region-labor ×1	DEFAULT

Confidence 80%: \$5.23–\$12.21/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

fdm / PLA — qty 1000 — \$8.68/unit

Line items

Cost element	USD/unit
labor	\$7.05
machine	\$1.41
material	\$0.09
amortized_fixed	\$0.13
Σ unit cost	\$8.68

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0927 \$	CAD volume 2.72 cm³ × PLA density 1.24 g/cm³ = 0.0034 kg × \$25/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1	DEFAULT
parts_per_build	68.0 parts	XY nest: plate 250×250mm × xy_packing 0.5 ÷ footprint (15.0×20.0+4mm) = 68 parts/plate	DEFAULT
machine_cost	1.4057 \$	0.1757 hr × \$8/hr × region-labor ×1 [serial XY-nested: deposition V/16 = 2.72/16 = 0.170hr/part (per-part nozzle) + Z-sweep (10.0/25)÷68 parts/plate = 0.006hr/part (plate Z-climb amortized; XY packing 0.5, plate 250×250mm) = 0.176hr/part]	DEFAULT
cycle_time	0.1757 hr	serial XY-nested: deposition V/16 = 2.72/16 = 0.170hr/part (per-part nozzle) + Z-sweep (10.0/25)÷68 parts/plate = 0.006hr/part (plate Z-climb amortized; XY packing 0.5, plate 250×250mm) = 0.176hr/part	DEFAULT
labor_cost	7.0515 \$	finish 0.2hr/part + bulk 0.1hr/build ÷ 68 = 0.201hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	0.1313 \$	setup 0.25hr × \$35/hr × ceil(1000/68) = 15 setups ÷ 1000 × region-labor ×1	DEFAULT

Confidence 80%: \$5.21–\$12.15/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

fdm / PLA — qty 2000 — \$8.68/unit

Line items

Cost element	USD/unit
labor	\$7.05
machine	\$1.41
material	\$0.09
amortized_fixed	\$0.13
Σ unit cost	\$8.68

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0927 \$	CAD volume 2.72 cm³ × PLA density 1.24 g/cm³ = 0.0034 kg × \$25/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1	DEFAULT
parts_per_build	68.0 parts	XY nest: plate 250×250mm × xy_packing 0.5 ÷ footprint (15.0×20.0+4mm) = 68 parts/plate	DEFAULT
machine_cost	1.4057 \$	0.1757 hr × \$8/hr × region-labor ×1 [serial XY-nested: deposition V/16 = 2.72/16 = 0.170hr/part (per-part nozzle) + Z-sweep (10.0/25)÷68 parts/plate = 0.006hr/part (plate Z-climb amortized; XY packing 0.5, plate 250×250mm) = 0.176hr/part]	DEFAULT
cycle_time	0.1757 hr	serial XY-nested: deposition V/16 = 2.72/16 = 0.170hr/part (per-part nozzle) + Z-sweep (10.0/25)÷68 parts/plate = 0.006hr/part (plate Z-climb amortized; XY packing 0.5, plate 250×250mm) = 0.176hr/part	DEFAULT
labor_cost	7.0515 \$	finish 0.2hr/part + bulk 0.1hr/build ÷ 68 = 0.201hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	0.1313 \$	setup 0.25hr × \$35/hr × ceil(2000/68) = 30 setups ÷ 2000 × region-labor ×1	DEFAULT

Confidence 80%: \$5.21–\$12.15/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

fdm / PLA — qty 5000 — \$8.68/unit

Line items

Cost element	USD/unit
labor	\$7.05
machine	\$1.41
material	\$0.09
amortized_fixed	\$0.13
Σ unit cost	\$8.68

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0927 \$	CAD volume 2.72 cm³ × PLA density 1.24 g/cm³ = 0.0034 kg × \$25/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1	DEFAULT
parts_per_build	68.0 parts	XY nest: plate 250×250mm × xy_packing 0.5 ÷ footprint (15.0×20.0+4mm) = 68 parts/plate	DEFAULT
machine_cost	1.4057 \$	0.1757 hr × \$8/hr × region-labor ×1 [serial XY-nested: deposition V/16 = 2.72/16 = 0.170hr/part (per-part nozzle) + Z-sweep (10.0/25)÷68 parts/plate = 0.006hr/part (plate Z-climb amortized; XY packing 0.5, plate 250×250mm) = 0.176hr/part]	DEFAULT
cycle_time	0.1757 hr	serial XY-nested: deposition V/16 = 2.72/16 = 0.170hr/part (per-part nozzle) + Z-sweep (10.0/25)÷68 parts/plate = 0.006hr/part (plate Z-climb amortized; XY packing 0.5, plate 250×250mm) = 0.176hr/part	DEFAULT
labor_cost	7.0515 \$	finish 0.2hr/part + bulk 0.1hr/build ÷ 68 = 0.201hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	0.1295 \$	setup 0.25hr × \$35/hr × ceil(5000/68) = 74 setups ÷ 5000 × region-labor ×1	DEFAULT

Confidence 80%: \$5.21–\$12.15/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

fdm / PLA — qty 10000 — \$8.68/unit

Line items

Cost element	USD/unit
labor	\$7.05
machine	\$1.41
material	\$0.09
amortized_fixed	\$0.13
Σ unit cost	\$8.68

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0927 \$	CAD volume 2.72 cm³ × PLA density 1.24 g/cm³ = 0.0034 kg × \$25/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1	DEFAULT
parts_per_build	68.0 parts	XY nest: plate 250×250mm × xy_packing 0.5 ÷ footprint (15.0×20.0+4mm) = 68 parts/plate	DEFAULT
machine_cost	1.4057 \$	0.1757 hr × \$8/hr × region-labor ×1 [serial XY-nested: deposition V/16 = 2.72/16 = 0.170hr/part (per-part nozzle) + Z-sweep (10.0/25)÷68 parts/plate = 0.006hr/part (plate Z-climb amortized; XY packing 0.5, plate 250×250mm) = 0.176hr/part]	DEFAULT
cycle_time	0.1757 hr	serial XY-nested: deposition V/16 = 2.72/16 = 0.170hr/part (per-part nozzle) + Z-sweep (10.0/25)÷68 parts/plate = 0.006hr/part (plate Z-climb amortized; XY packing 0.5, plate 250×250mm) = 0.176hr/part	DEFAULT
labor_cost	7.0515 \$	finish 0.2hr/part + bulk 0.1hr/build ÷ 68 = 0.201hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	0.1295 \$	setup 0.25hr × \$35/hr × ceil(10000/68) = 148 setups ÷ 10000 × region-labor ×1	DEFAULT

Confidence 80%: \$5.21–\$12.15/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

sla / Standard Resin — qty 1 — \$40.00/unit

Line items

Cost element	USD/unit
labor	\$5.53
machine	\$4.32
material	\$0.18
amortized_fixed	\$10.50
min_charge_floor	\$19.48
Σ unit cost	\$40.00

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.1764 \$	CAD volume 2.72 cm ³ × Standard Resin density 1.18 g/cm ³ = 0.0032 kg × \$50/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1	DEFAULT
parts_per_build	25.0 parts	XY nest: plate 145×145mm × xy_packing 0.5 ÷ footprint (15.0×20.0+3mm) = 25 parts/plate	DEFAULT
machine_cost	4.316 \$	0.3597 hr × \$12/hr × region-labor ×1 [serial XY-nested: deposition V/8 = 2.72/8 = 0.340hr/part (per-part nozzle) + Z-sweep (10.0/20)÷25 parts/plate = 0.020hr/part (plate Z-climb amortized; XY packing 0.5, plate 145×145mm) = 0.360hr/part]	DEFAULT
cycle_time	0.3597 hr	serial XY-nested: deposition V/8 = 2.72/8 = 0.340hr/part (per-part nozzle) + Z-sweep (10.0/20)÷25 parts/plate = 0.020hr/part (plate Z-climb amortized; XY packing 0.5, plate 145×145mm) = 0.360hr/part	DEFAULT
labor_cost	5.53 \$	finish 0.15hr/part + bulk 0.2hr/build ÷ 25 = 0.158hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	10.5 \$	setup 0.3hr × \$35/hr × ceil(1/25) = 1 setups ÷ 1 × region-labor ×1	DEFAULT
min_charge_floor	19.4776 \$	shop/order minimum \$40/lot × 1 lots ÷ 1 = \$40.00/unit floor (applied)	DEFAULT

Confidence 80%: \$24.00–\$56.00/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

sla / Standard Resin — qty 100 — \$10.44/unit

Line items

Cost element	USD/unit
labor	\$5.53
machine	\$4.32
material	\$0.18
amortized_fixed	\$0.42
Σ unit cost	\$10.44

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.1764 \$	CAD volume 2.72 cm³ × Standard Resin density 1.18 g/cm³ = 0.0032 kg × \$50/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1	DEFAULT
parts_per_build	25.0 parts	XY nest: plate 145×145mm × xy_packing 0.5 ÷ footprint (15.0×20.0+3mm) = 25 parts/plate	DEFAULT
machine_cost	4.316 \$	0.3597 hr × \$12/hr × region-labor ×1 [serial XY-nested: deposition V/8 = 2.72/8 = 0.340hr/part (per-part nozzle) + Z-sweep (10.0/20)÷25 parts/plate = 0.020hr/part (plate Z-climb amortized; XY packing 0.5, plate 145×145mm) = 0.360hr/part]	DEFAULT
cycle_time	0.3597 hr	serial XY-nested: deposition V/8 = 2.72/8 = 0.340hr/part (per-part nozzle) + Z-sweep (10.0/20)÷25 parts/plate = 0.020hr/part (plate Z-climb amortized; XY packing 0.5, plate 145×145mm) = 0.360hr/part	DEFAULT
labor_cost	5.53 \$	finish 0.15hr/part + bulk 0.2hr/build ÷ 25 = 0.158hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	0.42 \$	setup 0.3hr × \$35/hr × ceil(100/25) = 4 setups ÷ 100 × region-labor ×1	DEFAULT

Confidence 80%: \$6.27–\$14.62/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

sla / Standard Resin — qty 1000 — \$10.44/unit

Line items

Cost element	USD/unit
labor	\$5.53
machine	\$4.32
material	\$0.18
amortized_fixed	\$0.42
Σ unit cost	\$10.44

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.1764 \$	CAD volume 2.72 cm ³ × Standard Resin density 1.18 g/cm ³ = 0.0032 kg × \$50/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1	DEFAULT
parts_per_build	25.0 parts	XY nest: plate 145×145mm × xy_packing 0.5 ÷ footprint (15.0×20.0+3mm) = 25 parts/plate	DEFAULT
machine_cost	4.316 \$	0.3597 hr × \$12/hr × region-labor ×1 [serial XY-nested: deposition V/8 = 2.72/8 = 0.340hr/part (per-part nozzle) + Z-sweep (10.0/20)÷25 parts/plate = 0.020hr/part (plate Z-climb amortized; XY packing 0.5, plate 145×145mm) = 0.360hr/part]	DEFAULT
cycle_time	0.3597 hr	serial XY-nested: deposition V/8 = 2.72/8 = 0.340hr/part (per-part nozzle) + Z-sweep (10.0/20)÷25 parts/plate = 0.020hr/part (plate Z-climb amortized; XY packing 0.5, plate 145×145mm) = 0.360hr/part	DEFAULT
labor_cost	5.53 \$	finish 0.15hr/part + bulk 0.2hr/build ÷ 25 = 0.158hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	0.42 \$	setup 0.3hr × \$35/hr × ceil(1000/25) = 40 setups ÷ 1000 × region-labor ×1	DEFAULT

Confidence 80%: \$6.27–\$14.62/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

sla / Standard Resin — qty 2000 — \$10.44/unit

Line items

Cost element	USD/unit
labor	\$5.53
machine	\$4.32
material	\$0.18
amortized_fixed	\$0.42
Σ unit cost	\$10.44

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.1764 \$	CAD volume 2.72 cm³ × Standard Resin density 1.18 g/cm³ = 0.0032 kg × \$50/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1	DEFAULT
parts_per_build	25.0 parts	XY nest: plate 145×145mm × xy_packing 0.5 ÷ footprint (15.0×20.0+3mm) = 25 parts/plate	DEFAULT
machine_cost	4.316 \$	0.3597 hr × \$12/hr × region-labor ×1 [serial XY-nested: deposition V/8 = 2.72/8 = 0.340hr/part (per-part nozzle) + Z-sweep (10.0/20)÷25 parts/plate = 0.020hr/part (plate Z-climb amortized; XY packing 0.5, plate 145×145mm) = 0.360hr/part]	DEFAULT
cycle_time	0.3597 hr	serial XY-nested: deposition V/8 = 2.72/8 = 0.340hr/part (per-part nozzle) + Z-sweep (10.0/20)÷25 parts/plate = 0.020hr/part (plate Z-climb amortized; XY packing 0.5, plate 145×145mm) = 0.360hr/part	DEFAULT
labor_cost	5.53 \$	finish 0.15hr/part + bulk 0.2hr/build ÷ 25 = 0.158hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	0.42 \$	setup 0.3hr × \$35/hr × ceil(2000/25) = 80 setups ÷ 2000 × region-labor ×1	DEFAULT

Confidence 80%: \$6.27–\$14.62/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

sla / Standard Resin — qty 5000 — \$10.44/unit

Line items

Cost element	USD/unit
labor	\$5.53
machine	\$4.32
material	\$0.18
amortized_fixed	\$0.42
Σ unit cost	\$10.44

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.1764 \$	CAD volume 2.72 cm³ × Standard Resin density 1.18 g/cm³ = 0.0032 kg × \$50/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1	DEFAULT
parts_per_build	25.0 parts	XY nest: plate 145×145mm × xy_packing 0.5 ÷ footprint (15.0×20.0+3mm) = 25 parts/plate	DEFAULT
machine_cost	4.316 \$	0.3597 hr × \$12/hr × region-labor ×1 [serial XY-nested: deposition V/8 = 2.72/8 = 0.340hr/part (per-part nozzle) + Z-sweep (10.0/20)÷25 parts/plate = 0.020hr/part (plate Z-climb amortized; XY packing 0.5, plate 145×145mm) = 0.360hr/part]	DEFAULT
cycle_time	0.3597 hr	serial XY-nested: deposition V/8 = 2.72/8 = 0.340hr/part (per-part nozzle) + Z-sweep (10.0/20)÷25 parts/plate = 0.020hr/part (plate Z-climb amortized; XY packing 0.5, plate 145×145mm) = 0.360hr/part	DEFAULT
labor_cost	5.53 \$	finish 0.15hr/part + bulk 0.2hr/build ÷ 25 = 0.158hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	0.42 \$	setup 0.3hr × \$35/hr × ceil(5000/25) = 200 setups ÷ 5000 × region-labor ×1	DEFAULT

Confidence 80%: \$6.27–\$14.62/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

sla / Standard Resin — qty 10000 — \$10.44/unit

Line items

Cost element	USD/unit
labor	\$5.53
machine	\$4.32
material	\$0.18
amortized_fixed	\$0.42
Σ unit cost	\$10.44

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.1764 \$	CAD volume 2.72 cm ³ × Standard Resin density 1.18 g/cm ³ = 0.0032 kg × \$50/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1	DEFAULT
parts_per_build	25.0 parts	XY nest: plate 145×145mm × xy_packing 0.5 ÷ footprint (15.0×20.0+3mm) = 25 parts/plate	DEFAULT
machine_cost	4.316 \$	0.3597 hr × \$12/hr × region-labor ×1 [serial XY-nested: deposition V/8 = 2.72/8 = 0.340hr/part (per-part nozzle) + Z-sweep (10.0/20)÷25 parts/plate = 0.020hr/part (plate Z-climb amortized; XY packing 0.5, plate 145×145mm) = 0.360hr/part]	DEFAULT
cycle_time	0.3597 hr	serial XY-nested: deposition V/8 = 2.72/8 = 0.340hr/part (per-part nozzle) + Z-sweep (10.0/20)÷25 parts/plate = 0.020hr/part (plate Z-climb amortized; XY packing 0.5, plate 145×145mm) = 0.360hr/part	DEFAULT
labor_cost	5.53 \$	finish 0.15hr/part + bulk 0.2hr/build ÷ 25 = 0.158hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	0.42 \$	setup 0.3hr × \$35/hr × ceil(10000/25) = 400 setups ÷ 10000 × region-labor ×1	DEFAULT

Confidence 80%: \$6.27–\$14.62/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

sls / PA12 (Nylon 12) — qty 1 — \$75.00/unit

Line items

Cost element	USD/unit
labor	\$2.82
machine	\$0.65
material	\$0.18
amortized_fixed	\$17.50
min_charge_floor	\$53.85
Σ unit cost	\$75.00

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.1811 \$	CAD volume $2.72\text{ cm}^3 \times \text{PA12 (Nylon 12) density } 1.01\text{ g/cm}^3 = 0.0027\text{ kg} \times \$60/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.1\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
parts_per_build	924.0 parts	nesting: packing $0.1 \times \text{env } (340, 340, 600) \div \text{part bbox } (10.0, 15.0, 20.0)+5\text{mm}$ spacing = 924 parts/build	DEFAULT
machine_cost	0.6494 \$	$0.0325\text{ hr} \times \$20/\text{hr} \times \text{region-labor} \times 1$ [build-job 600mm $\div 20\text{mm/hr} = 30.0\text{hr}$ full build $\div 924\text{ parts/build}$ (packing 0.1, env (340, 340, 600)) = 0.032hr/part]	DEFAULT
cycle_time	0.0325 hr	build-job 600mm $\div 20\text{mm/hr} = 30.0\text{hr}$ full build $\div 924\text{ parts/build}$ (packing 0.1, env (340, 340, 600)) = 0.032hr/part	DEFAULT
labor_cost	2.8189 \$	finish $0.08\text{hr/part} + \text{bulk } 0.5\text{hr/build} \div 924 = 0.081\text{hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	17.5 \$	setup $0.5\text{hr} \times \$35/\text{hr} \times \text{ceil}(1/924) = 1\text{ setups} \div 1 \times \text{region-labor} \times 1$	DEFAULT
min_charge_floor	53.8506 \$	shop/order minimum $\$75/\text{lot} \times 1\text{ lots} \div 1 = \$75.00/\text{unit floor}$ (applied)	DEFAULT

Confidence 80%: \$45.00–\$105.00/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

sls / PA12 (Nylon 12) — qty 100 — \$3.82/unit

Line items

Cost element	USD/unit
labor	\$2.82
machine	\$0.65
material	\$0.18
amortized_fixed	\$0.17
Σ unit cost	\$3.82

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.1811 \$	CAD volume $2.72\text{ cm}^3 \times \text{PA12 (Nylon 12) density } 1.01\text{ g/cm}^3 = 0.0027\text{ kg} \times \$60/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.1\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
parts_per_build	924.0 parts	nesting: packing $0.1 \times \text{env} (340, 340, 600) \div \text{part bbox} (10.0, 15.0, 20.0)+5\text{mm}$ spacing = 924 parts/build	DEFAULT
machine_cost	0.6494 \$	$0.0325\text{ hr} \times \$20/\text{hr} \times \text{region-labor} \times 1$ [build-job $600\text{mm} \div 20\text{mm/hr} = 30.0\text{hr}$ full build $\div 924\text{ parts/build (packing } 0.1, \text{ env } (340, 340, 600)) = 0.032\text{hr/part}$]	DEFAULT
cycle_time	0.0325 hr	build-job $600\text{mm} \div 20\text{mm/hr} = 30.0\text{hr}$ full build $\div 924\text{ parts/build (packing } 0.1, \text{ env } (340, 340, 600)) = 0.032\text{hr/part}$	DEFAULT
labor_cost	2.8189 \$	finish $0.08\text{hr/part} + \text{bulk } 0.5\text{hr/build} \div 924 = 0.081\text{hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	0.175 \$	setup $0.5\text{hr} \times \$35/\text{hr} \times \text{ceil}(100/924) = 1\text{ setups} \div 100 \times \text{region-labor} \times 1$	DEFAULT

Confidence 80%: \$2.29–\$5.35/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

sls / PA12 (Nylon 12) — qty 1000 — \$3.68/unit

Line items

Cost element	USD/unit
labor	\$2.82
machine	\$0.65
material	\$0.18
amortized_fixed	\$0.04
Σ unit cost	\$3.68

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.1811 \$	CAD volume $2.72\text{ cm}^3 \times \text{PA12 (Nylon 12) density } 1.01\text{ g/cm}^3 = 0.0027\text{ kg} \times \$60/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.1\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
parts_per_build	924.0 parts	nesting: packing $0.1 \times \text{env} (340, 340, 600) \div \text{part bbox} (10.0, 15.0, 20.0)+5\text{mm}$ spacing = 924 parts/build	DEFAULT
machine_cost	0.6494 \$	$0.0325\text{ hr} \times \$20/\text{hr} \times \text{region-labor} \times 1$ [build-job 600mm $\div 20\text{mm/hr} = 30.0\text{hr}$ full build $\div 924\text{ parts/build}$ (packing 0.1, env (340, 340, 600)) = 0.032hr/part]	DEFAULT
cycle_time	0.0325 hr	build-job 600mm $\div 20\text{mm/hr} = 30.0\text{hr}$ full build $\div 924\text{ parts/build}$ (packing 0.1, env (340, 340, 600)) = 0.032hr/part	DEFAULT
labor_cost	2.8189 \$	finish $0.08\text{hr/part} + \text{bulk } 0.5\text{hr/build} \div 924 = 0.081\text{hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	0.035 \$	setup $0.5\text{hr} \times \$35/\text{hr} \times \text{ceil}(1000/924) = 2\text{ setups} \div 1000 \times \text{region-labor} \times 1$	DEFAULT

Confidence 80%: \$2.21–\$5.16/unit ($\pm 40.0\%$) — **assumption-based, not yet validated** · $\pm 40\%$ stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

sls / PA12 (Nylon 12) — qty 2000 — \$3.68/unit

Line items

Cost element	USD/unit
labor	\$2.82
machine	\$0.65
material	\$0.18
amortized_fixed	\$0.03
Σ unit cost	\$3.68

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.1811 \$	CAD volume $2.72\text{ cm}^3 \times \text{PA12 (Nylon 12) density } 1.01\text{ g/cm}^3 = 0.0027\text{ kg} \times \$60/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.1\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
parts_per_build	924.0 parts	nesting: packing $0.1 \times \text{env} (340, 340, 600) \div \text{part bbox} (10.0, 15.0, 20.0)+5\text{mm}$ spacing = 924 parts/build	DEFAULT
machine_cost	0.6494 \$	$0.0325\text{ hr} \times \$20/\text{hr} \times \text{region-labor} \times 1$ [build-job 600mm $\div 20\text{mm/hr} = 30.0\text{hr}$ full build $\div 924\text{ parts/build}$ (packing 0.1, env (340, 340, 600)) = 0.032hr/part]	DEFAULT
cycle_time	0.0325 hr	build-job 600mm $\div 20\text{mm/hr} = 30.0\text{hr}$ full build $\div 924\text{ parts/build}$ (packing 0.1, env (340, 340, 600)) = 0.032hr/part	DEFAULT
labor_cost	2.8189 \$	finish $0.08\text{hr/part} + \text{bulk } 0.5\text{hr/build} \div 924 = 0.081\text{hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	0.0262 \$	setup $0.5\text{hr} \times \$35/\text{hr} \times \text{ceil}(2000/924) = 3\text{ setups} \div 2000 \times \text{region-labor} \times 1$	DEFAULT

Confidence 80%: \$2.21–\$5.15/unit ($\pm 40.0\%$) — **assumption-based, not yet validated** · $\pm 40\%$ stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

sls / PA12 (Nylon 12) — qty 5000 — \$3.67/unit

Line items

Cost element	USD/unit
labor	\$2.82
machine	\$0.65
material	\$0.18
amortized_fixed	\$0.02
Σ unit cost	\$3.67

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.1811 \$	CAD volume $2.72\text{ cm}^3 \times \text{PA12 (Nylon 12) density } 1.01\text{ g/cm}^3 = 0.0027\text{ kg} \times \$60/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.1\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
parts_per_build	924.0 parts	nesting: packing $0.1 \times \text{env} (340, 340, 600) \div \text{part bbox} (10.0, 15.0, 20.0)+5\text{mm}$ spacing = 924 parts/build	DEFAULT
machine_cost	0.6494 \$	$0.0325\text{ hr} \times \$20/\text{hr} \times \text{region-labor} \times 1$ [build-job $600\text{mm} \div 20\text{mm/hr} = 30.0\text{hr}$ full build $\div 924\text{ parts/build (packing } 0.1, \text{ env } (340, 340, 600)) = 0.032\text{hr/part}$]	DEFAULT
cycle_time	0.0325 hr	build-job $600\text{mm} \div 20\text{mm/hr} = 30.0\text{hr}$ full build $\div 924\text{ parts/build (packing } 0.1, \text{ env } (340, 340, 600)) = 0.032\text{hr/part}$	DEFAULT
labor_cost	2.8189 \$	finish $0.08\text{hr/part} + \text{bulk } 0.5\text{hr/build} \div 924 = 0.081\text{hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	0.021 \$	setup $0.5\text{hr} \times \$35/\text{hr} \times \text{ceil}(5000/924) = 6\text{ setups} \div 5000 \times \text{region-labor} \times 1$	DEFAULT

Confidence 80%: \$2.20–\$5.14/unit ($\pm 40.0\%$) — **assumption-based, not yet validated** · $\pm 40\%$ stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

sls / PA12 (Nylon 12) — qty 10000 — \$3.67/unit

Line items

Cost element	USD/unit
labor	\$2.82
machine	\$0.65
material	\$0.18
amortized_fixed	\$0.02
Σ unit cost	\$3.67

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.1811 \$	CAD volume $2.72\text{ cm}^3 \times \text{PA12 (Nylon 12) density } 1.01\text{ g/cm}^3 = 0.0027\text{ kg} \times \$60/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.1\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
parts_per_build	924.0 parts	nesting: packing $0.1 \times \text{env} (340, 340, 600) \div \text{part bbox} (10.0, 15.0, 20.0)+5\text{mm}$ spacing = 924 parts/build	DEFAULT
machine_cost	0.6494 \$	$0.0325\text{ hr} \times \$20/\text{hr} \times \text{region-labor} \times 1$ [build-job $600\text{mm} \div 20\text{mm/hr} = 30.0\text{hr}$ full build $\div 924\text{ parts/build (packing } 0.1, \text{ env } (340, 340, 600)) = 0.032\text{hr/part}$]	DEFAULT
cycle_time	0.0325 hr	build-job $600\text{mm} \div 20\text{mm/hr} = 30.0\text{hr}$ full build $\div 924\text{ parts/build (packing } 0.1, \text{ env } (340, 340, 600)) = 0.032\text{hr/part}$	DEFAULT
labor_cost	2.8189 \$	finish $0.08\text{hr/part} + \text{bulk } 0.5\text{hr/build} \div 924 = 0.081\text{hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	0.0192 \$	setup $0.5\text{hr} \times \$35/\text{hr} \times \text{ceil}(10000/924) = 11\text{ setups} \div 10000 \times \text{region-labor} \times 1$	DEFAULT

Confidence 80%: \$2.20–\$5.14/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

cnc_5axis / Delrin (POM) — qty 1 — \$188.85/unit

Line items

Cost element	USD/unit
nre	\$105.00
labor	\$17.50
machine	\$3.17
material	\$0.02
inspection	\$28.00
consumables	\$0.16
amortized_fixed	\$35.00
Σ unit cost	\$188.85

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0244 \$	bounding-box billet $3.00\text{ cm}^3 \times 1.10$ stock allowance $\times$ Delrin (POM) density 1.41 g/cm^3 [assumption, not shop-validated] = $0.0047\text{ kg} \times \$5/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.05\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
machine_cost	3.1717 \$	$0.0288\text{ hr} \times \$110/\text{hr} \times \text{region-labor} \times 1$ [rough $0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr} + \text{finish } 14.3\text{ cm}^2 \div 500\text{ cm}^2/\text{hr} = 0.029\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]]	DEFAULT
cycle_time	0.0288 hr	$\text{rough } 0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr} + \text{finish } 14.3\text{ cm}^2 \div 500\text{ cm}^2/\text{hr} = 0.029\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]	DEFAULT
labor_cost	17.5 \$	post-process $0.5\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	35.0 \$	setup $1\text{ hr} \times \$35/\text{hr} \times \text{ceil}(1/100) = 1\text{ setups} \div 1 \times \text{region-labor} \times 1$	DEFAULT
nre_cost	105.0 \$	CAM programming/NRE $3\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1 \div 1\text{ order (one-time)}$ [assumption, not shop-validated]	DEFAULT
inspection_cost	28.0 \$	first-article $0.75\text{ hr} \times 1\text{ lot(s)} \div 1 + \text{in-process } 0.05\text{ hr}/\text{part} = 0.8000\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$ [assumption, not shop-validated]	DEFAULT
consumables_cost	0.1586 \$	perishable tooling/consumables $0.05 \times \text{machine } \3.17 (cutting tools, inserts, coolant) [assumption, not shop-validated]	DEFAULT

Confidence 80%: \$94.43–\$283.28/unit (±50.0%) — **assumption-based, not yet validated** · ±50% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

cnc_5axis / Delrin (POM) — qty 100 — \$24.27/unit

Line items

Cost element	USD/unit
nre	\$1.05
labor	\$17.50
machine	\$3.17
material	\$0.02
inspection	\$2.01
consumables	\$0.16
amortized_fixed	\$0.35
Σ unit cost	\$24.27

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0244 \$	bounding-box billet $3.00\text{ cm}^3 \times 1.10$ stock allowance $\times$ Delrin (POM) density 1.41 g/cm^3 [assumption, not shop-validated] = $0.0047\text{ kg} \times \$5/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.05\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
machine_cost	3.1717 \$	$0.0288\text{ hr} \times \$110/\text{hr} \times \text{region-labor} \times 1$ [rough $0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr} + \text{finish } 14.3\text{ cm}^2 \div 500\text{ cm}^2/\text{hr} = 0.029\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]]	DEFAULT
cycle_time	0.0288 hr	$\text{rough } 0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr} + \text{finish } 14.3\text{ cm}^2 \div 500\text{ cm}^2/\text{hr} = 0.029\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]	DEFAULT
labor_cost	17.5 \$	post-process $0.5\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	0.35 \$	setup $1\text{ hr} \times \$35/\text{hr} \times \text{ceil}(100/100) = 1\text{ setups} \div 100 \times \text{region-labor} \times 1$	DEFAULT
nre_cost	1.05 \$	CAM programming/NRE $3\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1 \div 100$ order (one-time) [assumption, not shop-validated]	DEFAULT
inspection_cost	2.0125 \$	first-article $0.75\text{ hr} \times 1\text{ lot(s)} \div 100 + \text{in-process } 0.05\text{ hr/part} = 0.0575\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$ [assumption, not shop-validated]	DEFAULT
consumables_cost	0.1586 \$	perishable tooling/consumables $0.05 \times \text{machine } \3.17 (cutting tools, inserts, coolant) [assumption, not shop-validated]	DEFAULT

Confidence 80%: \$12.13–\$36.40/unit (±50.0%) — **assumption-based, not yet validated** · ±50% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

cnc_5axis / Delrin (POM) — qty 1000 — \$16.65/unit

Line items

Cost element	USD/unit
nre	\$0.10
labor	\$12.33
machine	\$2.23
material	\$0.02
inspection	\$1.50
consumables	\$0.11
amortized_fixed	\$0.35
Σ unit cost	\$16.65

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0244 \$	bounding-box billet $3.00\text{ cm}^3 \times 1.10$ stock allowance $\times$ Delrin (POM) density 1.41 g/cm^3 [assumption, not shop-validated] = $0.0047\text{ kg} \times \$5/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.05\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
machine_cost	2.235 \$	$0.0288\text{ hr} \times \$110/\text{hr} \times 0.705\text{ learning@qty1000} \times \text{region-labor} \times 1$ [rough $0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 500\text{ cm}^2/\text{hr} = 0.029\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]]	DEFAULT
cycle_time	0.0288 hr	$\text{rough } 0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 500\text{ cm}^2/\text{hr} = 0.029\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]	DEFAULT
labor_cost	12.332 \$	post-process $0.5\text{ hr} \times \$35/\text{hr} \times 0.705\text{ learning@qty1000} \times \text{region-labor} \times 1$	DEFAULT
learning_curve	0.7047 x	learning curve $0.9 \times / \text{doubling of cumulative qty on machine+labor: } (1000/100\text{ first-lot})^{-0.152} = \times 0.705$ (~3.3 doublings, floor 0.25) [assumption, not shop-validated]	DEFAULT
setup_cost	0.35 \$	setup $1\text{ hr} \times \$35/\text{hr} \times \text{ceil}(1000/100) = 10\text{ setups} \div 1000 \times \text{region-labor} \times 1$	DEFAULT
nre_cost	0.105 \$	CAM programming/NRE $3\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1 \div 1000\text{ order (one-time)}$ [assumption, not shop-validated]	DEFAULT
inspection_cost	1.4957 \$	first-article $0.75\text{ hr} \times 10\text{ lot(s)} \div 1000$ + in-process $0.05\text{ hr/part} \times 0.705\text{ learning@qty1000} = 0.0427\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$ [assumption, not shop-validated]	DEFAULT
consumables_cost	0.1118 \$	perishable tooling/consumables $0.05 \times \text{machine } \2.24 (cutting tools, inserts, coolant) [assumption, not shop-validated]	DEFAULT

Confidence 80%: \$8.33–\$24.98/unit (±50.0%) — **assumption-based, not yet validated** · ±50% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

cnc_5axis / Delrin (POM) — qty 2000 — \$15.01/unit

Line items

Cost element	USD/unit
nre	\$0.05
labor	\$11.10
machine	\$2.01
material	\$0.02
inspection	\$1.37
consumables	\$0.10
amortized_fixed	\$0.35
Σ unit cost	\$15.01

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0244 \$	bounding-box billet $3.00\text{ cm}^3 \times 1.10$ stock allowance $\times$ Delrin (POM) density 1.41 g/cm^3 [assumption, not shop-validated] = $0.0047\text{ kg} \times \$5/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.05\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
machine_cost	2.0115 \$	$0.0288\text{ hr} \times \$110/\text{hr} \times 0.634\text{ learning@qty2000} \times \text{region-labor} \times 1$ [rough $0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 500\text{ cm}^2/\text{hr} = 0.029\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]]	DEFAULT
cycle_time	0.0288 hr	rough $0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 500\text{ cm}^2/\text{hr} = 0.029\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]	DEFAULT
labor_cost	11.0988 \$	post-process $0.5\text{ hr} \times \$35/\text{hr} \times 0.634\text{ learning@qty2000} \times \text{region-labor} \times 1$	DEFAULT
learning_curve	0.6342 $\times$	learning curve $0.9 \times \text{doubling of cumulative qty on machine+labor: } (2000/100\text{ first-lot})^{-0.152} = \times 0.634$ (~4.3 doublings, floor 0.25) [assumption, not shop-validated]	DEFAULT
setup_cost	0.35 \$	setup $1\text{ hr} \times \$35/\text{hr} \times \text{ceil}(2000/100) = 20\text{ setups} \div 2000 \times \text{region-labor} \times 1$	DEFAULT
nre_cost	0.0525 \$	CAM programming/NRE $3\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1 \div 2000\text{ order (one-time)}$ [assumption, not shop-validated]	DEFAULT
inspection_cost	1.3724 \$	first-article $0.75\text{ hr} \times 20\text{ lot(s)} \div 2000$ + in-process $0.05\text{ hr/part} \times 0.634\text{ learning@qty2000} = 0.0392\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$ [assumption, not shop-validated]	DEFAULT
consumables_cost	0.1006 \$	perishable tooling/consumables $0.05 \times \text{machine } \2.01 (cutting tools, inserts, coolant) [assumption, not shop-validated]	DEFAULT

Confidence 80%: \$7.51–\$22.52/unit ($\pm 50.0\%$) — **assumption-based, not yet validated** · $\pm 50\%$ stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

cnc_5axis / Delrin (POM) — qty 5000 — \$13.12/unit

Line items

Cost element	USD/unit
nre	\$0.02
labor	\$9.66
machine	\$1.75
material	\$0.02
inspection	\$1.23
consumables	\$0.09
amortized_fixed	\$0.35
Σ unit cost	\$13.12

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0244 \$	bounding-box billet $3.00\text{ cm}^3 \times 1.10$ stock allowance $\times$ Delrin (POM) density 1.41 g/cm^3 [assumption, not shop-validated] = $0.0047\text{ kg} \times \$5/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.05\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
machine_cost	1.75 \$	$0.0288\text{ hr} \times \$110/\text{hr} \times 0.552\text{ learning@qty5000} \times \text{region-labor} \times 1$ [rough $0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 500\text{ cm}^2/\text{hr} = 0.029\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]]	DEFAULT
cycle_time	0.0288 hr	$\text{rough } 0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 500\text{ cm}^2/\text{hr} = 0.029\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]	DEFAULT
labor_cost	9.6558 \$	post-process $0.5\text{ hr} \times \$35/\text{hr} \times 0.552\text{ learning@qty5000} \times \text{region-labor} \times 1$	DEFAULT
learning_curve	0.5518 x	learning curve $0.9 \times / \text{doubling of cumulative qty on machine+labor: } (5000/100\text{ first-lot})^{-0.152} = \times 0.552$ (~5.6 doublings, floor 0.25) [assumption, not shop-validated]	DEFAULT
setup_cost	0.35 \$	setup $1\text{ hr} \times \$35/\text{hr} \times \text{ceil}(5000/100) = 50\text{ setups} \div 5000 \times \text{region-labor} \times 1$	DEFAULT
nre_cost	0.021 \$	CAM programming/NRE $3\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1 \div 5000\text{ order (one-time)}$ [assumption, not shop-validated]	DEFAULT
inspection_cost	1.2281 \$	first-article $0.75\text{ hr} \times 50\text{ lot(s)} \div 5000$ + in-process $0.05\text{ hr/part} \times 0.552\text{ learning@qty5000} = 0.0351\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$ [assumption, not shop-validated]	DEFAULT
consumables_cost	0.0875 \$	perishable tooling/consumables $0.05 \times \text{machine } \1.75 (cutting tools, inserts, coolant) [assumption, not shop-validated]	DEFAULT

Confidence 80%: \$6.56–\$19.68/unit (±50.0%) — **assumption-based, not yet validated** · ±50% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

cnc_5axis / Delrin (POM) — qty 10000 — \$11.86/unit

Line items

Cost element	USD/unit
nre	\$0.01
labor	\$8.69
machine	\$1.57
material	\$0.02
inspection	\$1.13
consumables	\$0.08
amortized_fixed	\$0.35
Σ unit cost	\$11.86

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0244 \$	bounding-box billet $3.00\text{ cm}^3 \times 1.10$ stock allowance $\times$ Delrin (POM) density 1.41 g/cm^3 [assumption, not shop-validated] = $0.0047\text{ kg} \times \$5/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.05\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
machine_cost	1.575 \$	$0.0288\text{ hr} \times \$110/\text{hr} \times 0.497\text{ learning@qty10000} \times \text{region-labor} \times 1$ [rough $0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 500\text{ cm}^2/\text{hr} = 0.029\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]]	DEFAULT
cycle_time	0.0288 hr	rough $0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 500\text{ cm}^2/\text{hr} = 0.029\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]	DEFAULT
labor_cost	8.6902 \$	post-process $0.5\text{ hr} \times \$35/\text{hr} \times 0.497\text{ learning@qty10000} \times \text{region-labor} \times 1$	DEFAULT
learning_curve	0.4966 x	learning curve $0.9 \times / \text{doubling of cumulative qty on machine+labor: } (10000/100\text{ first-lot})^{-0.152} = \times 0.497$ (~6.6 doublings, floor 0.25) [assumption, not shop-validated]	DEFAULT
setup_cost	0.35 \$	setup $1\text{ hr} \times \$35/\text{hr} \times \text{ceil}(10000/100) = 100\text{ setups} \div 10000 \times \text{region-labor} \times 1$	DEFAULT
nre_cost	0.0105 \$	CAM programming/NRE $3\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1 \div 10000\text{ order (one-time)}$ [assumption, not shop-validated]	DEFAULT
inspection_cost	1.1315 \$	first-article $0.75\text{ hr} \times 100\text{ lot(s)} \div 10000$ + in-process $0.05\text{ hr/part} \times 0.497\text{ learning@qty10000} = 0.0323\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$ [assumption, not shop-validated]	DEFAULT
consumables_cost	0.0788 \$	perishable tooling/consumables $0.05 \times \text{machine } \1.58 (cutting tools, inserts, coolant) [assumption, not shop-validated]	DEFAULT

Confidence 80%: \$5.93–\$17.79/unit (±50.0%) — **assumption-based, not yet validated** · ±50% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

mjf / PP (Polypropylene) — qty 1 — \$75.00/unit

Line items

Cost element	USD/unit
labor	\$2.83
machine	\$0.61
material	\$0.01
amortized_fixed	\$17.50
min_charge_floor	\$54.05
Σ unit cost	\$75.00

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0054 \$	CAD volume $2.72\text{ cm}^3 \times \text{PP (Polypropylene) density } 0.90\text{ g/cm}^3 = 0.0024\text{ kg} \times \$2/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.1\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
parts_per_build	546.0 parts	nesting: $\text{packing } 0.1 \times \text{env } (380, 284, 380) \div \text{part bbox } (10.0, 15.0, 20.0)+5\text{mm spacing} = 546\text{ parts/build}$	DEFAULT
machine_cost	0.6125 \$	$0.0278\text{ hr} \times \$22/\text{hr} \times \text{region-labor} \times 1$ [build-job 380mm $\div 25\text{mm/hr} = 15.2\text{hr full build} \div 546\text{ parts/build (packing } 0.1, \text{env } (380, 284, 380)) = 0.028\text{hr/part}$]	DEFAULT
cycle_time	0.0278 hr	build-job 380mm $\div 25\text{mm/hr} = 15.2\text{hr full build} \div 546\text{ parts/build (packing } 0.1, \text{env } (380, 284, 380)) = 0.028\text{hr/part}$	DEFAULT
labor_cost	2.8321 \$	finish $0.08\text{hr/part} + \text{bulk } 0.5\text{hr/build} \div 546 = 0.081\text{hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	17.5 \$	setup $0.5\text{hr} \times \$35/\text{hr} \times \text{ceil}(1/546) = 1\text{ setups} \div 1 \times \text{region-labor} \times 1$	DEFAULT
min_charge_floor	54.05 \$	shop/order minimum $\$75/\text{lot} \times 1\text{ lots} \div 1 = \$75.00/\text{unit floor (applied)}$	DEFAULT

Confidence 80%: \$45.00–\$105.00/unit ($\pm 40.0\%$) — **assumption-based, not yet validated** · $\pm 40\%$ stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

mjf / PP (Polypropylene) — qty 100 — \$3.62/unit

Line items

Cost element	USD/unit
labor	\$2.83
machine	\$0.61
material	\$0.01
amortized_fixed	\$0.17
Σ unit cost	\$3.62

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0054 \$	CAD volume $2.72\text{ cm}^3 \times \text{PP (Polypropylene) density } 0.90\text{ g/cm}^3 = 0.0024\text{ kg} \times \$2/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.1\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
parts_per_build	546.0 parts	nesting: packing $0.1 \times \text{env } (380, 284, 380) \div \text{part bbox } (10.0, 15.0, 20.0)+5\text{mm}$ spacing = 546 parts/build	DEFAULT
machine_cost	0.6125 \$	$0.0278\text{ hr} \times \$22/\text{hr} \times \text{region-labor} \times 1$ [build-job $380\text{mm} \div 25\text{mm/hr} = 15.2\text{hr}$ full build $\div 546\text{ parts/build}$ (packing 0.1, env (380, 284, 380)) = 0.028hr/part]	DEFAULT
cycle_time	0.0278 hr	build-job $380\text{mm} \div 25\text{mm/hr} = 15.2\text{hr}$ full build $\div 546\text{ parts/build}$ (packing 0.1, env (380, 284, 380)) = 0.028hr/part	DEFAULT
labor_cost	2.8321 \$	finish $0.08\text{hr/part} + \text{bulk } 0.5\text{hr/build} \div 546 = 0.081\text{hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	0.175 \$	setup $0.5\text{hr} \times \$35/\text{hr} \times \text{ceil}(100/546) = 1\text{ setups} \div 100 \times \text{region-labor} \times 1$	DEFAULT

Confidence 80%: \$2.17–\$5.07/unit ($\pm 40.0\%$) — **assumption-based, not yet validated** · $\pm 40\%$ stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

mjf / PP (Polypropylene) — qty 1000 — \$3.48/unit

Line items

Cost element	USD/unit
labor	\$2.83
machine	\$0.61
material	\$0.01
amortized_fixed	\$0.04
Σ unit cost	\$3.48

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0054 \$	CAD volume $2.72\text{ cm}^3 \times \text{PP (Polypropylene) density } 0.90\text{ g/cm}^3 = 0.0024\text{ kg} \times \$2/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.1\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
parts_per_build	546.0 parts	nesting: packing $0.1 \times \text{env } (380, 284, 380) \div \text{part bbox } (10.0, 15.0, 20.0)+5\text{mm}$ spacing = 546 parts/build	DEFAULT
machine_cost	0.6125 \$	$0.0278\text{ hr} \times \$22/\text{hr} \times \text{region-labor} \times 1$ [build-job $380\text{mm} \div 25\text{mm/hr} = 15.2\text{hr}$ full build $\div 546\text{ parts/build}$ (packing 0.1, env (380, 284, 380)) = 0.028hr/part]	DEFAULT
cycle_time	0.0278 hr	build-job $380\text{mm} \div 25\text{mm/hr} = 15.2\text{hr}$ full build $\div 546\text{ parts/build}$ (packing 0.1, env (380, 284, 380)) = 0.028hr/part	DEFAULT
labor_cost	2.8321 \$	finish $0.08\text{hr/part} + \text{bulk } 0.5\text{hr/build} \div 546 = 0.081\text{hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	0.035 \$	setup $0.5\text{hr} \times \$35/\text{hr} \times \text{ceil}(1000/546) = 2\text{ setups} \div 1000 \times \text{region-labor} \times 1$	DEFAULT

Confidence 80%: \$2.09–\$4.88/unit ($\pm 40.0\%$) — **assumption-based, not yet validated** · $\pm 40\%$ stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

mjf / PP (Polypropylene) — qty 2000 — \$3.48/unit

Line items

Cost element	USD/unit
labor	\$2.83
machine	\$0.61
material	\$0.01
amortized_fixed	\$0.04
Σ unit cost	\$3.48

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0054 \$	CAD volume 2.72 cm³ × PP (Polypropylene) density 0.90 g/cm³ = 0.0024 kg × \$2/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1	DEFAULT
parts_per_build	546.0 parts	nesting: packing 0.1 × env (380, 284, 380) ÷ part bbox (10.0, 15.0, 20.0)+5mm spacing = 546 parts/build	DEFAULT
machine_cost	0.6125 \$	0.0278 hr × \$22/hr × region-labor ×1 [build-job 380mm ÷ 25mm/hr = 15.2hr full build ÷ 546 parts/build (packing 0.1, env (380, 284, 380)) = 0.028hr/part]	DEFAULT
cycle_time	0.0278 hr	build-job 380mm ÷ 25mm/hr = 15.2hr full build ÷ 546 parts/build (packing 0.1, env (380, 284, 380)) = 0.028hr/part	DEFAULT
labor_cost	2.8321 \$	finish 0.08hr/part + bulk 0.5hr/build ÷ 546 = 0.081hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	0.035 \$	setup 0.5hr × \$35/hr × ceil(2000/546) = 4 setups ÷ 2000 × region-labor ×1	DEFAULT

Confidence 80%: \$2.09–\$4.88/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

mjf / PP (Polypropylene) — qty 5000 — \$3.48/unit

Line items

Cost element	USD/unit
labor	\$2.83
machine	\$0.61
material	\$0.01
amortized_fixed	\$0.04
Σ unit cost	\$3.48

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0054 \$	CAD volume $2.72\text{ cm}^3 \times \text{PP (Polypropylene) density } 0.90\text{ g/cm}^3 = 0.0024\text{ kg} \times \$2/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.1\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
parts_per_build	546.0 parts	nesting: packing $0.1 \times \text{env } (380, 284, 380) \div \text{part bbox } (10.0, 15.0, 20.0)+5\text{mm}$ spacing = 546 parts/build	DEFAULT
machine_cost	0.6125 \$	$0.0278\text{ hr} \times \$22/\text{hr} \times \text{region-labor} \times 1$ [build-job $380\text{mm} \div 25\text{mm/hr} = 15.2\text{hr}$ full build $\div 546\text{ parts/build}$ (packing 0.1, env (380, 284, 380)) = 0.028hr/part]	DEFAULT
cycle_time	0.0278 hr	build-job $380\text{mm} \div 25\text{mm/hr} = 15.2\text{hr}$ full build $\div 546\text{ parts/build}$ (packing 0.1, env (380, 284, 380)) = 0.028hr/part	DEFAULT
labor_cost	2.8321 \$	finish $0.08\text{hr/part} + \text{bulk } 0.5\text{hr/build} \div 546 = 0.081\text{hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	0.035 \$	setup $0.5\text{hr} \times \$35/\text{hr} \times \text{ceil}(5000/546) = 10\text{ setups} \div 5000 \times \text{region-labor} \times 1$	DEFAULT

Confidence 80%: \$2.09–\$4.88/unit ($\pm 40.0\%$) — **assumption-based, not yet validated** · $\pm 40\%$ stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

mjf / PP (Polypropylene) — qty 10000 — \$3.48/unit

Line items

Cost element	USD/unit
labor	\$2.83
machine	\$0.61
material	\$0.01
amortized_fixed	\$0.03
Σ unit cost	\$3.48

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0054 \$	CAD volume $2.72\text{ cm}^3 \times \text{PP (Polypropylene) density } 0.90\text{ g/cm}^3 = 0.0024\text{ kg} \times \$2/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.1\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
parts_per_build	546.0 parts	nesting: packing $0.1 \times \text{env } (380, 284, 380) \div \text{part bbox } (10.0, 15.0, 20.0)+5\text{mm}$ spacing = 546 parts/build	DEFAULT
machine_cost	0.6125 \$	$0.0278\text{ hr} \times \$22/\text{hr} \times \text{region-labor} \times 1$ [build-job 380mm $\div 25\text{mm/hr} = 15.2\text{hr}$ full build $\div 546\text{ parts/build}$ (packing 0.1, env (380, 284, 380)) = 0.028hr/part]	DEFAULT
cycle_time	0.0278 hr	build-job 380mm $\div 25\text{mm/hr} = 15.2\text{hr}$ full build $\div 546\text{ parts/build}$ (packing 0.1, env (380, 284, 380)) = 0.028hr/part	DEFAULT
labor_cost	2.8321 \$	finish $0.08\text{hr/part} + \text{bulk } 0.5\text{hr/build} \div 546 = 0.081\text{hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	0.0333 \$	setup $0.5\text{hr} \times \$35/\text{hr} \times \text{ceil}(10000/546) = 19\text{ setups} \div 10000 \times \text{region-labor} \times 1$	DEFAULT

Confidence 80%: \$2.09–\$4.88/unit ($\pm 40.0\%$) — **assumption-based, not yet validated** · $\pm 40\%$ stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

cnc_3axis / Delrin (POM) — qty 1 — \$134.22/unit

Line items

Cost element	USD/unit
nre	\$70.00
labor	\$17.50
machine	\$1.80
material	\$0.02
inspection	\$18.55
consumables	\$0.09
amortized_fixed	\$26.25
Σ unit cost	\$134.22

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0244 \$	bounding-box billet $3.00\text{ cm}^3 \times 1.10$ stock allowance $\times$ Delrin (POM) density 1.41 g/cm^3 [assumption, not shop-validated] = $0.0047\text{ kg} \times \$5/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.05\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
machine_cost	1.8045 \$	$0.0241\text{ hr} \times \$75/\text{hr} \times \text{region-labor} \times 1$ [rough $0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 600\text{ cm}^2/\text{hr} = 0.024\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]]	DEFAULT
cycle_time	0.0241 hr	rough $0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 600\text{ cm}^2/\text{hr} = 0.024\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]	DEFAULT
labor_cost	17.5 \$	post-process $0.5\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	26.25 \$	setup $0.75\text{ hr} \times \$35/\text{hr} \times \text{ceil}(1/100) = 1\text{ setups} \div 1 \times \text{region-labor} \times 1$	DEFAULT
nre_cost	70.0 \$	CAM programming/NRE $2\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1 \div 1\text{ order (one-time)}$ [assumption, not shop-validated]	DEFAULT
inspection_cost	18.55 \$	first-article $0.5\text{ hr} \times 1\text{ lot(s)} \div 1 + \text{in-process } 0.03\text{ hr/part} = 0.5300\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$ [assumption, not shop-validated]	DEFAULT
consumables_cost	0.0902 \$	perishable tooling/consumables $0.05 \times \text{machine } \1.80 (cutting tools, inserts, coolant) [assumption, not shop-validated]	DEFAULT

Confidence 80%: \$67.11–\$201.33/unit (±50.0%) — **assumption-based, not yet validated** · ±50% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

cnc_3axis / Delrin (POM) — qty 100 — \$21.61/unit

Line items

Cost element	USD/unit
nre	\$0.70
labor	\$17.50
machine	\$1.80
material	\$0.02
inspection	\$1.23
consumables	\$0.09
amortized_fixed	\$0.26
Σ unit cost	\$21.61

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0244 \$	bounding-box billet $3.00\text{ cm}^3 \times 1.10$ stock allowance $\times$ Delrin (POM) density 1.41 g/cm^3 [assumption, not shop-validated] = $0.0047\text{ kg} \times \$5/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.05\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
machine_cost	1.8045 \$	$0.0241\text{ hr} \times \$75/\text{hr} \times \text{region-labor} \times 1$ [rough $0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 600\text{ cm}^2/\text{hr} = 0.024\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]]	DEFAULT
cycle_time	0.0241 hr	rough $0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 600\text{ cm}^2/\text{hr} = 0.024\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]	DEFAULT
labor_cost	17.5 \$	post-process $0.5\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	0.2625 \$	setup $0.75\text{ hr} \times \$35/\text{hr} \times \text{ceil}(100/100) = 1\text{ setups} \div 100 \times \text{region-labor} \times 1$	DEFAULT
nre_cost	0.7 \$	CAM programming/NRE $2\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1 \div 100$ order (one-time) [assumption, not shop-validated]	DEFAULT
inspection_cost	1.225 \$	first-article $0.5\text{ hr} \times 1\text{ lot(s)} \div 100$ + in-process $0.03\text{ hr/part} = 0.0350\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$ [assumption, not shop-validated]	DEFAULT
consumables_cost	0.0902 \$	perishable tooling/consumables $0.05 \times \text{machine } \1.80 (cutting tools, inserts, coolant) [assumption, not shop-validated]	DEFAULT

Confidence 80%: \$10.80–\$32.41/unit (±50.0%) — **assumption-based, not yet validated** · ±50% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

cnc_3axis / Delrin (POM) — qty 1000 — \$14.94/unit

Line items

Cost element	USD/unit
nre	\$0.07
labor	\$12.33
machine	\$1.27
material	\$0.02
inspection	\$0.91
consumables	\$0.06
amortized_fixed	\$0.26
Σ unit cost	\$14.94

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0244 \$	bounding-box billet $3.00\text{ cm}^3 \times 1.10$ stock allowance $\times$ Delrin (POM) density 1.41 g/cm^3 [assumption, not shop-validated] = $0.0047\text{ kg} \times \$5/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.05\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
machine_cost	1.2716 \$	$0.0241\text{ hr} \times \$75/\text{hr} \times 0.705\text{ learning@qty1000} \times \text{region-labor} \times 1$ [rough $0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 600\text{ cm}^2/\text{hr} = 0.024\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]]	DEFAULT
cycle_time	0.0241 hr	$\text{rough } 0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 600\text{ cm}^2/\text{hr} = 0.024\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]	DEFAULT
labor_cost	12.332 \$	post-process $0.5\text{ hr} \times \$35/\text{hr} \times 0.705\text{ learning@qty1000} \times \text{region-labor} \times 1$	DEFAULT
learning_curve	0.7047 x	learning curve $0.9 \times / \text{doubling of cumulative qty on machine+labor: } (1000/100\text{ first-lot})^{-0.152} = \times 0.705$ (~3.3 doublings, floor 0.25) [assumption, not shop-validated]	DEFAULT
setup_cost	0.2625 \$	setup $0.75\text{ hr} \times \$35/\text{hr} \times \text{ceil}(1000/100) = 10\text{ setups} \div 1000 \times \text{region-labor} \times 1$	DEFAULT
nre_cost	0.07 \$	CAM programming/NRE $2\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1 \div 1000$ order (one-time) [assumption, not shop-validated]	DEFAULT
inspection_cost	0.9149 \$	first-article $0.5\text{ hr} \times 10\text{ lot(s)} \div 1000$ + in-process $0.03\text{ hr/part} \times 0.705\text{ learning@qty1000} = 0.0261\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$ [assumption, not shop-validated]	DEFAULT
consumables_cost	0.0636 \$	perishable tooling/consumables $0.05 \times \text{machine } \1.27 (cutting tools, inserts, coolant) [assumption, not shop-validated]	DEFAULT

Confidence 80%: \$7.47–\$22.41/unit (±50.0%) — **assumption-based, not yet validated** · ±50% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

cnc_3axis / Delrin (POM) — qty 2000 — \$13.46/unit

Line items

Cost element	USD/unit
nre	\$0.04
labor	\$11.10
machine	\$1.14
material	\$0.02
inspection	\$0.84
consumables	\$0.06
amortized_fixed	\$0.26
Σ unit cost	\$13.46

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0244 \$	bounding-box billet $3.00\text{ cm}^3 \times 1.10$ stock allowance $\times$ Delrin (POM) density 1.41 g/cm^3 [assumption, not shop-validated] = $0.0047\text{ kg} \times \$5/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.05\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
machine_cost	1.1445 \$	$0.0241\text{ hr} \times \$75/\text{hr} \times 0.634\text{ learning@qty2000} \times \text{region-labor} \times 1$ [rough $0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 600\text{ cm}^2/\text{hr} = 0.024\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]]	DEFAULT
cycle_time	0.0241 hr	$\text{rough } 0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 600\text{ cm}^2/\text{hr} = 0.024\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]	DEFAULT
labor_cost	11.0988 \$	post-process $0.5\text{ hr} \times \$35/\text{hr} \times 0.634\text{ learning@qty2000} \times \text{region-labor} \times 1$	DEFAULT
learning_curve	0.6342 $\times$	learning curve $0.9 \times \text{doubling of cumulative qty on machine+labor: } (2000/100\text{ first-lot})^{-0.152} = \times 0.634$ (~4.3 doublings, floor 0.25) [assumption, not shop-validated]	DEFAULT
setup_cost	0.2625 \$	setup $0.75\text{ hr} \times \$35/\text{hr} \times \text{ceil}(2000/100) = 20\text{ setups} \div 2000 \times \text{region-labor} \times 1$	DEFAULT
nre_cost	0.035 \$	CAM programming/NRE $2\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1 \div 2000\text{ order (one-time)}$ [assumption, not shop-validated]	DEFAULT
inspection_cost	0.8409 \$	first-article $0.5\text{ hr} \times 20\text{ lot(s)} \div 2000$ + in-process $0.03\text{ hr/part} \times 0.634\text{ learning@qty2000} = 0.0240\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$ [assumption, not shop-validated]	DEFAULT
consumables_cost	0.0572 \$	perishable tooling/consumables $0.05 \times \text{machine } \1.14 (cutting tools, inserts, coolant) [assumption, not shop-validated]	DEFAULT

Confidence 80%: \$6.73–\$20.19/unit (±50.0%) — **assumption-based, not yet validated** · ±50% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

cnc_3axis / Delrin (POM) — qty 5000 — \$11.76/unit

Line items

Cost element	USD/unit
nre	\$0.01
labor	\$9.66
machine	\$1.00
material	\$0.02
inspection	\$0.75
consumables	\$0.05
amortized_fixed	\$0.26
Σ unit cost	\$11.76

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0244 \$	bounding-box billet $3.00\text{ cm}^3 \times 1.10$ stock allowance $\times$ Delrin (POM) density 1.41 g/cm^3 [assumption, not shop-validated] = $0.0047\text{ kg} \times \$5/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.05\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
machine_cost	0.9957 \$	$0.0241\text{ hr} \times \$75/\text{hr} \times 0.552\text{ learning@qty5000} \times \text{region-labor} \times 1$ [rough $0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 600\text{ cm}^2/\text{hr} = 0.024\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]]	DEFAULT
cycle_time	0.0241 hr	$\text{rough } 0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 600\text{ cm}^2/\text{hr} = 0.024\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]	DEFAULT
labor_cost	9.6558 \$	post-process $0.5\text{ hr} \times \$35/\text{hr} \times 0.552\text{ learning@qty5000} \times \text{region-labor} \times 1$	DEFAULT
learning_curve	0.5518 $\times$	learning curve $0.9 \times / \text{doubling of cumulative qty on machine+labor: } (5000/100\text{ first-lot})^{-0.152} = \times 0.552$ (~5.6 doublings, floor 0.25) [assumption, not shop-validated]	DEFAULT
setup_cost	0.2625 \$	setup $0.75\text{ hr} \times \$35/\text{hr} \times \text{ceil}(5000/100) = 50\text{ setups} \div 5000 \times \text{region-labor} \times 1$	DEFAULT
nre_cost	0.014 \$	CAM programming/NRE $2\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1 \div 5000$ order (one-time) [assumption, not shop-validated]	DEFAULT
inspection_cost	0.7543 \$	first-article $0.5\text{ hr} \times 50\text{ lot(s)} \div 5000$ + in-process $0.03\text{ hr/part} \times 0.552\text{ learning@qty5000} = 0.0216\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$ [assumption, not shop-validated]	DEFAULT
consumables_cost	0.0498 \$	perishable tooling/consumables $0.05 \times \text{machine } \1.00 (cutting tools, inserts, coolant) [assumption, not shop-validated]	DEFAULT

Confidence 80%: \$5.88–\$17.63/unit (±50.0%) — **assumption-based, not yet validated** · ±50% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

cnc_3axis / Delrin (POM) — qty 10000 — \$10.62/unit

Line items

Cost element	USD/unit
nre	\$0.01
labor	\$8.69
machine	\$0.90
material	\$0.02
inspection	\$0.70
consumables	\$0.04
amortized_fixed	\$0.26
Σ unit cost	\$10.62

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.0244 \$	bounding-box billet $3.00\text{ cm}^3 \times 1.10$ stock allowance $\times$ Delrin (POM) density 1.41 g/cm^3 [assumption, not shop-validated] = $0.0047\text{ kg} \times \$5/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.05\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
machine_cost	0.8961 \$	$0.0241\text{ hr} \times \$75/\text{hr} \times 0.497\text{ learning@qty10000} \times \text{region-labor} \times 1$ [rough $0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 600\text{ cm}^2/\text{hr} = 0.024\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]]	DEFAULT
cycle_time	0.0241 hr	$\text{rough } 0.6\text{ cm}^3 \div (50\text{ cm}^3/\text{min} \cdot 60) = 0.000\text{ hr}$ + finish $14.3\text{ cm}^2 \div 600\text{ cm}^2/\text{hr} = 0.024\text{ hr}$ [stock bbox billet $3.0\text{ cm}^3 \times 1.10 = 3.3\text{ cm}^3$ [assumption, not shop-validated]; MRR polymer]	DEFAULT
labor_cost	8.6902 \$	post-process $0.5\text{ hr} \times \$35/\text{hr} \times 0.497\text{ learning@qty10000} \times \text{region-labor} \times 1$	DEFAULT
learning_curve	0.4966 $\times$	learning curve $0.9 \times / \text{doubling of cumulative qty on machine+labor: } (10000/100\text{ first-lot})^{-0.152} = \times 0.497$ (~6.6 doublings, floor 0.25) [assumption, not shop-validated]	DEFAULT
setup_cost	0.2625 \$	setup $0.75\text{ hr} \times \$35/\text{hr} \times \text{ceil}(10000/100) = 100\text{ setups} \div 10000 \times \text{region-labor} \times 1$	DEFAULT
nre_cost	0.007 \$	CAM programming/NRE $2\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1 \div 10000$ order (one-time) [assumption, not shop-validated]	DEFAULT
inspection_cost	0.6964 \$	first-article $0.5\text{ hr} \times 100\text{ lot(s)} \div 10000$ + in-process $0.03\text{ hr/part} \times 0.497\text{ learning@qty10000} = 0.0199\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$ [assumption, not shop-validated]	DEFAULT
consumables_cost	0.0448 \$	perishable tooling/consumables $0.05 \times \text{machine } \0.90 (cutting tools, inserts, coolant) [assumption, not shop-validated]	DEFAULT

Confidence 80%: \$5.31–\$15.93/unit (±50.0%) — **assumption-based, not yet validated** · ±50% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

injection_molding / PP (Molded) — qty 1 — \$6,002.18/unit ⚠ not DFM-ready as-modeled

Line items

Cost element	USD/unit
labor	\$1.75
machine	\$0.42
material	\$0.01
amortized_fixed	\$6,000.00
Σ unit cost	\$6,002.18

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.005 \$	CAD volume 2.72 cm^3 × PP (Molded) density 0.90 g/cm^3 = 0.0024 kg × \$2/kg (material-DB unit price (DEFAULT book value)) × (1+0.03 scrap) × region-material ×1	DEFAULT
machine_cost	0.4226 \$	$0.0094\text{ hr} \times \$45/\text{hr} \times \text{region-labor} \times 1$ [cooling $2 \cdot \text{wall}^2 = 2 \cdot 3.80^2 = 28.8\text{ s}$ + shot 5s = 33.8s = 0.0094 hr [wall = 2V/A proxy, ±50%]]	DEFAULT
cycle_time	0.0094 hr	cooling $2 \cdot \text{wall}^2 = 2 \cdot 3.80^2 = 28.8\text{ s}$ + shot 5s = 33.8s = 0.0094 hr [wall = 2V/A proxy, ±50%]	DEFAULT
labor_cost	1.75 \$	post-process $0.05\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	0.0 \$	setup $0\text{ hr} \times \$35/\text{hr} \times \text{ceil}(1/100000) = 1\text{ setups} \div 1 \times \text{region-labor} \times 1$	DEFAULT
tooling_cost	6000.0 \$	size tier S (max bbox 20mm) × 1 cav ^{0.7} (=1.00) × moderate (=1.00) = \$6,000; ±60%, OVERRIDABLE	DEFAULT

Confidence 80%: \$2,400.87–\$9,603.48/unit (±60.0%) — **assumption-based, not yet validated** · ±60% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

injection_molding / PP (Molded) — qty 100 — \$62.18/unit ⚠ not DFM-ready as-modeled

Line items

Cost element	USD/unit
labor	\$1.75
machine	\$0.42
material	\$0.01
amortized_fixed	\$60.00
Σ unit cost	\$62.18

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.005 \$	CAD volume $2.72\text{ cm}^3 \times \text{PP (Molded) density } 0.90\text{ g/cm}^3 = 0.0024\text{ kg} \times \$2/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.03\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
machine_cost	0.4226 \$	$0.0094\text{ hr} \times \$45/\text{hr} \times \text{region-labor} \times 1$ [cooling $2 \cdot \text{wall}^2 = 2 \cdot 3.80^2 = 28.8\text{s} + \text{shot } 5\text{s} = 33.8\text{s} = 0.0094\text{ hr}$ [wall = $2V/A$ proxy, $\pm 50\%$]]	DEFAULT
cycle_time	0.0094 hr	cooling $2 \cdot \text{wall}^2 = 2 \cdot 3.80^2 = 28.8\text{s} + \text{shot } 5\text{s} = 33.8\text{s} = 0.0094\text{ hr}$ [wall = $2V/A$ proxy, $\pm 50\%$]	DEFAULT
labor_cost	1.75 \$	post-process $0.05\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	0.0 \$	setup $0\text{hr} \times \$35/\text{hr} \times \text{ceil}(100/100000) = 1\text{ setups} \div 100 \times \text{region-labor} \times 1$	DEFAULT
tooling_cost	6000.0 \$	size tier S (max bbox 20mm) $\times 1\text{ cav}^{0.7} (=1.00) \times \text{moderate} (=1.00) = \$6,000$; $\pm 60\%$, OVERRIDABLE	DEFAULT

Confidence 80%: \$24.87–\$99.48/unit ($\pm 60.0\%$) — **assumption-based, not yet validated** · $\pm 60\%$ stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

injection_molding / PP (Molded) — qty 1000 — \$8.18/unit ⚠ not DFM-ready as-modeled

Line items

Cost element	USD/unit
labor	\$1.75
machine	\$0.42
material	\$0.01
amortized_fixed	\$6.00
Σ unit cost	\$8.18

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.005 \$	CAD volume 2.72 cm³ × PP (Molded) density 0.90 g/cm³ = 0.0024 kg × \$2/kg (material-DB unit price (DEFAULT book value)) × (1+0.03 scrap) × region-material ×1	DEFAULT
machine_cost	0.4226 \$	0.0094 hr × \$45/hr × region-labor ×1 [cooling 2·wall² = 2·3.80² = 28.8s + shot 5s = 33.8s = 0.0094 hr [wall = 2V/A proxy, ±50%]]	DEFAULT
cycle_time	0.0094 hr	cooling 2·wall² = 2·3.80² = 28.8s + shot 5s = 33.8s = 0.0094 hr [wall = 2V/A proxy, ±50%]	DEFAULT
labor_cost	1.75 \$	post-process 0.05 hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	0.0 \$	setup 0hr × \$35/hr × ceil(1000/100000) = 1 setups ÷ 1000 × region-labor ×1	DEFAULT
tooling_cost	6000.0 \$	size tier S (max bbox 20mm) × 1 cav^0.7 (=1.00) × moderate (=1.00) = \$6,000; ±60%, OVERRIDABLE	DEFAULT

Confidence 80%: \$3.27–\$13.08/unit (±60.0%) — **assumption-based, not yet validated** · ±60% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

injection_molding / PP (Molded) — qty 2000 — \$5.18/unit ⚠ not DFM-ready as-modeled

Line items

Cost element	USD/unit
labor	\$1.75
machine	\$0.42
material	\$0.01
amortized_fixed	\$3.00
Σ unit cost	\$5.18

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.005 \$	CAD volume 2.72 cm^3 × PP (Molded) density 0.90 g/cm^3 = 0.0024 kg × \$2/kg (material-DB unit price (DEFAULT book value)) × (1+0.03 scrap) × region-material ×1	DEFAULT
machine_cost	0.4226 \$	$0.0094\text{ hr} \times \$45/\text{hr} \times \text{region-labor} \times 1$ [cooling $2 \cdot \text{wall}^2 = 2 \cdot 3.80^2 = 28.8\text{ s}$ + shot 5s = 33.8s = 0.0094 hr [wall = $2V/A$ proxy, ±50%]]	DEFAULT
cycle_time	0.0094 hr	cooling $2 \cdot \text{wall}^2 = 2 \cdot 3.80^2 = 28.8\text{ s}$ + shot 5s = 33.8s = 0.0094 hr [wall = $2V/A$ proxy, ±50%]	DEFAULT
labor_cost	1.75 \$	post-process $0.05\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	0.0 \$	setup $0\text{ hr} \times \$35/\text{hr} \times \text{ceil}(2000/100000) = 1\text{ setups} \div 2000 \times \text{region-labor} \times 1$	DEFAULT
tooling_cost	6000.0 \$	size tier S (max bbox 20mm) × 1 $\text{cav}^{0.7} (=1.00)$ × moderate (=1.00) = \$6,000; ±60%, OVERRIDABLE	DEFAULT

Confidence 80%: \$2.07–\$8.28/unit (±60.0%) — **assumption-based, not yet validated** · ±60% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

injection_molding / PP (Molded) — qty 5000 — \$3.38/unit ⚠ not DFM-ready as-modeled

Line items

Cost element	USD/unit
labor	\$1.75
machine	\$0.42
material	\$0.01
amortized_fixed	\$1.20
Σ unit cost	\$3.38

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.005 \$	CAD volume $2.72\text{ cm}^3 \times \text{PP (Molded) density } 0.90\text{ g/cm}^3 = 0.0024\text{ kg} \times \$2/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.03\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
machine_cost	0.4226 \$	$0.0094\text{ hr} \times \$45/\text{hr} \times \text{region-labor} \times 1$ [cooling $2 \cdot \text{wall}^2 = 2 \cdot 3.80^2 = 28.8\text{ s} + \text{shot } 5\text{ s} = 33.8\text{ s} = 0.0094\text{ hr}$ [wall = $2V/A$ proxy, $\pm 50\%$]]	DEFAULT
cycle_time	0.0094 hr	cooling $2 \cdot \text{wall}^2 = 2 \cdot 3.80^2 = 28.8\text{ s} + \text{shot } 5\text{ s} = 33.8\text{ s} = 0.0094\text{ hr}$ [wall = $2V/A$ proxy, $\pm 50\%$]	DEFAULT
labor_cost	1.75 \$	post-process $0.05\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	0.0 \$	setup $0\text{ hr} \times \$35/\text{hr} \times \text{ceil}(5000/100000) = 1\text{ setups} \div 5000 \times \text{region-labor} \times 1$	DEFAULT
tooling_cost	6000.0 \$	size tier S (max bbox 20mm) $\times 1\text{ cav}^{0.7} (=1.00) \times \text{moderate} (=1.00) = \$6,000$; $\pm 60\%$, OVERRIDABLE	DEFAULT

Confidence 80%: \$1.35–\$5.40/unit ($\pm 60.0\%$) — **assumption-based, not yet validated** · $\pm 60\%$ stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

injection_molding / PP (Molded) — qty 10000 — \$2.78/unit ⚠ **not DFM-ready as-modeled**

Line items

Cost element	USD/unit
labor	\$1.75
machine	\$0.42
material	\$0.01
amortized_fixed	\$0.60
Σ unit cost	\$2.78

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.005 \$	CAD volume 2.72 cm^3 × PP (Molded) density 0.90 g/cm^3 = 0.0024 kg × \$2/kg (material-DB unit price (DEFAULT book value)) × (1+0.03 scrap) × region-material ×1	DEFAULT
machine_cost	0.4226 \$	$0.0094\text{ hr} \times \$45/\text{hr} \times \text{region-labor} \times 1$ [cooling $2 \cdot \text{wall}^2 = 2 \cdot 3.80^2 = 28.8\text{ s}$ + shot 5s = 33.8s = 0.0094 hr [wall = 2V/A proxy, ±50%]]	DEFAULT
cycle_time	0.0094 hr	cooling $2 \cdot \text{wall}^2 = 2 \cdot 3.80^2 = 28.8\text{ s}$ + shot 5s = 33.8s = 0.0094 hr [wall = 2V/A proxy, ±50%]	DEFAULT
labor_cost	1.75 \$	post-process $0.05\text{ hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	0.0 \$	setup $0\text{ hr} \times \$35/\text{hr} \times \text{ceil}(10000/100000) = 1\text{ setups} \div 10000 \times \text{region-labor} \times 1$	DEFAULT
tooling_cost	6000.0 \$	size tier S (max bbox 20mm) × 1 $\text{cav}^{0.7} (=1.00)$ × moderate (=1.00) = \$6,000; ±60%, OVERRIDABLE	DEFAULT

Confidence 80%: \$1.11–\$4.44/unit (±60.0%) — **assumption-based, not yet validated** · ±60% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

dlp / Standard Resin — qty 1 — \$40.00/unit

Line items

Cost element	USD/unit
labor	\$5.31
machine	\$0.78
material	\$0.18
amortized_fixed	\$10.50
min_charge_floor	\$23.23
Σ unit cost	\$40.00

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.1764 \$	CAD volume 2.72 cm ³ × Standard Resin density 1.18 g/cm ³ = 0.0032 kg × \$50/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1	DEFAULT
parts_per_build	125.0 parts	nesting: packing 0.12 × env (192, 120, 245) ÷ part bbox (10.0, 15.0, 20.0)+3mm spacing = 125 parts/build	DEFAULT
machine_cost	0.784 \$	0.0653 hr × \$12/hr × region-labor ×1 [build-job 245mm ÷ 30mm/hr = 8.2hr full build ÷ 125 parts/build (packing 0.12, env (192, 120, 245)) = 0.065hr/part]	DEFAULT
cycle_time	0.0653 hr	build-job 245mm ÷ 30mm/hr = 8.2hr full build ÷ 125 parts/build (packing 0.12, env (192, 120, 245)) = 0.065hr/part	DEFAULT
labor_cost	5.306 \$	finish 0.15hr/part + bulk 0.2hr/build ÷ 125 = 0.152hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	10.5 \$	setup 0.3hr × \$35/hr × ceil(1/125) = 1 setups ÷ 1 × region-labor ×1	DEFAULT
min_charge_floor	23.2336 \$	shop/order minimum \$40/lot × 1 lots ÷ 1 = \$40.00/unit floor (applied)	DEFAULT

Confidence 80%: \$24.00–\$56.00/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

dlp / Standard Resin — qty 100 — \$6.37/unit

Line items

Cost element	USD/unit
labor	\$5.31
machine	\$0.78
material	\$0.18
amortized_fixed	\$0.10
Σ unit cost	\$6.37

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.1764 \$	CAD volume 2.72 cm ³ × Standard Resin density 1.18 g/cm ³ = 0.0032 kg × \$50/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1	DEFAULT
parts_per_build	125.0 parts	nesting: packing 0.12 × env (192, 120, 245) ÷ part bbox (10.0, 15.0, 20.0)+3mm spacing = 125 parts/build	DEFAULT
machine_cost	0.784 \$	0.0653 hr × \$12/hr × region-labor ×1 [build-job 245mm ÷ 30mm/hr = 8.2hr full build ÷ 125 parts/build (packing 0.12, env (192, 120, 245)) = 0.065hr/part]	DEFAULT
cycle_time	0.0653 hr	build-job 245mm ÷ 30mm/hr = 8.2hr full build ÷ 125 parts/build (packing 0.12, env (192, 120, 245)) = 0.065hr/part	DEFAULT
labor_cost	5.306 \$	finish 0.15hr/part + bulk 0.2hr/build ÷ 125 = 0.152hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	0.105 \$	setup 0.3hr × \$35/hr × ceil(100/125) = 1 setups ÷ 100 × region-labor ×1	DEFAULT

Confidence 80%: \$3.82–\$8.92/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

dlp / Standard Resin — qty 1000 — \$6.35/unit

Line items

Cost element	USD/unit
labor	\$5.31
machine	\$0.78
material	\$0.18
amortized_fixed	\$0.08
Σ unit cost	\$6.35

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.1764 \$	CAD volume 2.72 cm³ × Standard Resin density 1.18 g/cm³ = 0.0032 kg × \$50/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1	DEFAULT
parts_per_build	125.0 parts	nesting: packing 0.12 × env (192, 120, 245) ÷ part bbox (10.0, 15.0, 20.0)+3mm spacing = 125 parts/build	DEFAULT
machine_cost	0.784 \$	0.0653 hr × \$12/hr × region-labor ×1 [build-job 245mm ÷ 30mm/hr = 8.2hr full build ÷ 125 parts/build (packing 0.12, env (192, 120, 245)) = 0.065hr/part]	DEFAULT
cycle_time	0.0653 hr	build-job 245mm ÷ 30mm/hr = 8.2hr full build ÷ 125 parts/build (packing 0.12, env (192, 120, 245)) = 0.065hr/part	DEFAULT
labor_cost	5.306 \$	finish 0.15hr/part + bulk 0.2hr/build ÷ 125 = 0.152hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	0.084 \$	setup 0.3hr × \$35/hr × ceil(1000/125) = 8 setups ÷ 1000 × region-labor ×1	DEFAULT

Confidence 80%: \$3.81–\$8.89/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

dlp / Standard Resin — qty 2000 — \$6.35/unit

Line items

Cost element	USD/unit
labor	\$5.31
machine	\$0.78
material	\$0.18
amortized_fixed	\$0.08
Σ unit cost	\$6.35

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.1764 \$	CAD volume $2.72\text{ cm}^3 \times \text{Standard Resin density } 1.18\text{ g/cm}^3 = 0.0032\text{ kg} \times \$50/\text{kg}$ (material-DB unit price (DEFAULT book value)) $\times (1+0.1\text{ scrap}) \times \text{region-material} \times 1$	DEFAULT
parts_per_build	125.0 parts	nesting: packing $0.12 \times \text{env} (192, 120, 245) \div \text{part bbox} (10.0, 15.0, 20.0)+3\text{mm}$ spacing = 125 parts/build	DEFAULT
machine_cost	0.784 \$	$0.0653\text{ hr} \times \$12/\text{hr} \times \text{region-labor} \times 1$ [build-job $245\text{mm} \div 30\text{mm/hr} = 8.2\text{hr full build} \div 125\text{ parts/build}$ (packing 0.12, env (192, 120, 245)) = 0.065hr/part]	DEFAULT
cycle_time	0.0653 hr	build-job $245\text{mm} \div 30\text{mm/hr} = 8.2\text{hr full build} \div 125\text{ parts/build}$ (packing 0.12, env (192, 120, 245)) = 0.065hr/part	DEFAULT
labor_cost	5.306 \$	finish $0.15\text{hr/part} + \text{bulk } 0.2\text{hr/build} \div 125 = 0.152\text{hr} \times \$35/\text{hr} \times \text{region-labor} \times 1$	DEFAULT
setup_cost	0.084 \$	setup $0.3\text{hr} \times \$35/\text{hr} \times \text{ceil}(2000/125) = 16\text{ setups} \div 2000 \times \text{region-labor} \times 1$	DEFAULT

Confidence 80%: \$3.81–\$8.89/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

dlp / Standard Resin — qty 5000 — \$6.35/unit

Line items

Cost element	USD/unit
labor	\$5.31
machine	\$0.78
material	\$0.18
amortized_fixed	\$0.08
Σ unit cost	\$6.35

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.1764 \$	CAD volume 2.72 cm ³ × Standard Resin density 1.18 g/cm ³ = 0.0032 kg × \$50/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1	DEFAULT
parts_per_build	125.0 parts	nesting: packing 0.12 × env (192, 120, 245) ÷ part bbox (10.0, 15.0, 20.0)+3mm spacing = 125 parts/build	DEFAULT
machine_cost	0.784 \$	0.0653 hr × \$12/hr × region-labor ×1 [build-job 245mm ÷ 30mm/hr = 8.2hr full build ÷ 125 parts/build (packing 0.12, env (192, 120, 245)) = 0.065hr/part]	DEFAULT
cycle_time	0.0653 hr	build-job 245mm ÷ 30mm/hr = 8.2hr full build ÷ 125 parts/build (packing 0.12, env (192, 120, 245)) = 0.065hr/part	DEFAULT
labor_cost	5.306 \$	finish 0.15hr/part + bulk 0.2hr/build ÷ 125 = 0.152hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	0.084 \$	setup 0.3hr × \$35/hr × ceil(5000/125) = 40 setups ÷ 5000 × region-labor ×1	DEFAULT

Confidence 80%: \$3.81–\$8.89/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

dlp / Standard Resin — qty 10000 — \$6.35/unit

Line items

Cost element	USD/unit
labor	\$5.31
machine	\$0.78
material	\$0.18
amortized_fixed	\$0.08
Σ unit cost	\$6.35

Cost drivers & provenance

Driver	Value	Source	Provenance
material_cost	0.1764 \$	CAD volume 2.72 cm³ × Standard Resin density 1.18 g/cm³ = 0.0032 kg × \$50/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1	DEFAULT
parts_per_build	125.0 parts	nesting: packing 0.12 × env (192, 120, 245) ÷ part bbox (10.0, 15.0, 20.0)+3mm spacing = 125 parts/build	DEFAULT
machine_cost	0.784 \$	0.0653 hr × \$12/hr × region-labor ×1 [build-job 245mm ÷ 30mm/hr = 8.2hr full build ÷ 125 parts/build (packing 0.12, env (192, 120, 245)) = 0.065hr/part]	DEFAULT
cycle_time	0.0653 hr	build-job 245mm ÷ 30mm/hr = 8.2hr full build ÷ 125 parts/build (packing 0.12, env (192, 120, 245)) = 0.065hr/part	DEFAULT
labor_cost	5.306 \$	finish 0.15hr/part + bulk 0.2hr/build ÷ 125 = 0.152hr × \$35/hr × region-labor ×1	DEFAULT
setup_cost	0.084 \$	setup 0.3hr × \$35/hr × ceil(10000/125) = 80 setups ÷ 10000 × region-labor ×1	DEFAULT

Confidence 80%: \$3.81–\$8.89/unit (±40.0%) — **assumption-based, not yet validated** · ±40% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

Make-vs-Buy Crossover

Make now: fdm / PLA · **Tooling / production candidate:** injection_molding (redesign required)

Crossover quantity: ≈ 923.0 units — above this the tooling/production route becomes the cheaper per-unit choice.

Make by fdm (PLA) — \$30.0/unit at qty 1, the cheapest make-as-is option and your low-volume pick. fdm stays cheapest up to ~923 units; above ~923, injection_molding is cheaper (\$2.78/unit at qty 10000). Note: injection_molding requires design-for-molding — the part currently FAILS draft DFM (114818 sidewall faces (100.0% of sidewall area) below 1.0° draft for injection_molding.); the tooling cost shown is 'if redesigned for molding', not a current-capability quote.

Recommendation by quantity

Quantity	Process	Material	Unit cost	Cheaper if redesigned
1	fdm	PLA	\$30.00/unit	—
100	mjf	PP (Polypropylene)	\$3.62/unit	—
1000	mjf	PP (Polypropylene)	\$3.48/unit	—
2000	mjf	PP (Polypropylene)	\$3.48/unit	—
5000	mjf	PP (Polypropylene)	\$3.48/unit	injection_molding \$3.38/unit (add draft, tooling-dominated)
10000	mjf	PP (Polypropylene)	\$3.48/unit	injection_molding \$2.78/unit (add draft, tooling-dominated)

Assumptions Log

DEFAULT = generic fallback · SHOP = your calibrated profile · USER = ad-hoc override. Every assumption is editable and re-costable.

Assumption	Value	Unit	Source	Provenance
labor_rate	35.0	\$/hr	loaded shop-floor labor	DEFAULT
region_labor	1.0	×	region US: labor/machine/setup ×1	DEFAULT
region_material	1.0	×	region US: commodity material ×1 (global feedstock, ~1.0)	DEFAULT
region_tooling	1.0	×	region US: tooling ×1 (offshore toolmaking labor)	DEFAULT
margin	0.0	frac	target margin (price vs should-cost)	DEFAULT
overhead	0.0	frac	indirect burden on conversion cost (machine+labor+setup)	DEFAULT
utilization	1.0	frac	machine utilization (idle-recovery on machine cost)	DEFAULT
stock_allowance	1.1	×	CNC billet oversize (milling: on bounding box; turning: on hull)	DEFAULT
machine_labor_frac	0.35	frac	operator-labor share of the machine rate that scales with region (capital/facility stays global) [assumption, not shop-validated]	DEFAULT
perishable_frac	0.05	frac	perishable tooling/consumables as a fraction of CNC machine cost [assumption, not shop-validated]	DEFAULT
daily_machine_hours	8.0	hr/day	for lead-time production days	DEFAULT
n_cavities	1.0	cav	formative tooling cavities = 1 (DEFAULT single-cavity should-cost)	DEFAULT
complexity	0.0	moderate	tooling complexity = moderate (DEFAULT moderate; raise for slides/side-actions)	DEFAULT
material_class	0.0	polymer	material class = polymer (DEFAULT: these are polymer automotive parts; override for metal)	DEFAULT

Notes

- DFM note: processes injection_molding are costed but flagged NOT DFM-ready as-modeled (the engine reports ERROR-level blockers — these parts were modeled for 3D printing, so molding/casting lack draft). Their cost shows the tooling economics *if* the part is redesigned for that process. Set strict_dfm=True to exclude them entirely.
- Geometric routing: this part reads as a 'bulk_solid' → mjf. General solid (91% of bbox filled, ~3.8mm mean wall) with no dominant sheet/rotational/prismatic signature → additive (MJF/SLS) for a polymer prototype.
- Note: the cost-cheapest make-as-is pick at low qty is fdm, but the geometry-recommended process is mjf — both are costed above; pick on intent, not just the marginal \$.
- Absolute cost is ±40–60% (cycle-time/tooling defaults). The crossover quantity and make-vs-buy direction are robust to it because they depend on the fixed-vs-variable split, driven by your rates.